CSTE4103 — Cloud Computing Lab

Lab 2: Hosting Specs and Pricing

Read a server specification, then estimate the real monthly cost on AWS, Azure, and Google Cloud

Week 2 • Cybersecurity Engineering — 4th Year

Dr. Ruaa Ali Khamees • ICT Department
Polytechnic College for Engineering Specializations – Baghdad, Middle Technical University

Objectives and Deliverable

By the end you can

- Read a hosting specification: vCPU, memory, storage, network, region
- Match a workload to an instance family
- Use the official pricing calculators of three providers
- Spot hidden costs that raise the final bill
- Judge an offer on security and compliance, not price alone

Your deliverable

- One comparison table (monthly and yearly cost) for the same workload on 3 providers
- A short recommendation: 5 lines, with a security justification
- Note the date you took the prices, because they change

Anatomy of a Hosting Specification

vCPU

Virtual processor cores. More cores help parallel work; one busy core limits single-threaded apps.

Memory (RAM)

Working memory. Databases and caches are hungry for it; running out causes slowdowns.

Storage

SSD or HDD, size, and IOPS (operations per second). Fast disks cost more.

Network

Bandwidth and, importantly, outbound data transfer, which is usually billed.

Region and zone

Where the server physically lives: affects latency, price, and legal data residency.

OS and licensing

Linux is usually cheaper; Windows or SQL Server licenses add to the hourly price.

Instance Families: Match the Workload

Family	Good for	AWS	Azure	Google Cloud
General purpose	Web servers, small databases, dev and test	M, T	D, B	E2, N2
Compute optimized	Batch processing, encoding, gaming servers	C	F	C2
Memory optimized	In-memory caches, large databases, analytics	R	E	M2
Storage optimized	High-IOPS databases, data warehouses	I, D	L	Z3
GPU / accelerated	Machine learning, rendering	P, G	N	A2, G2

Burstable instances (such as AWS T or Azure B) are cheap but throttle under sustained load; dedicated cores cost more but stay predictable. Series letters are examples; names evolve.

Scenario: A University Student Portal

A department needs to host a small student portal (web app plus database). Price the same requirement on all three providers.

2	4 GB	50 GB	500 GB	Linux	730 h
vCPU	Memory	SSD storage	Outbound / month	OS	Running 24/7

Rules

- Pick one region close to Iraq's users, and use the same region logic on all three providers
- Choose the cheapest instance that still meets the requirement (not the biggest)
- Use on-demand pricing for the main table; note any discount for a 1-year commitment

Using the Official Pricing Calculators

AWS

Service: EC2 instance

calculator.aws

Azure

Service: Virtual Machines

azure.microsoft.com/pricing/calculator

Google Cloud

Service: Compute Engine

cloud.google.com/products/calculator

1. Open the calculator and add the compute service
2. Select the region
3. Choose the instance that matches 2 vCPU / 4 GB
4. Add 50 GB SSD storage
5. Add 500 GB outbound data transfer
6. Record the monthly total and the date

Hidden Costs That Change the Bill

The instance price is only the start. Check each of these before you trust a quote.

Data egress

Traffic leaving the cloud is billed; ingress is usually free.

Public IP addresses

Some providers charge for static or in-use public IPs.

Snapshots and backups

Every backup consumes billed storage.

Load balancer

Billed per hour plus per data processed.

Monitoring and logs

Log ingestion and retention add up quickly.

Support plans

Business-grade support is often a percentage of spend.

Licenses

Windows and database licenses raise hourly prices.

Idle resources

A forgotten VM or disk keeps billing until deleted.

Comparison Table to Fill In

Item	AWS	Azure	Google Cloud
Region chosen			
Instance name			
vCPU / RAM			
Compute per month			
Storage per month			
Data transfer per month			
Total per month			
Total per year (×12)			
1-year commitment discount			

Date prices were taken: ____ / ____ / _____ *Prices change often, so always cite the date.*

5 — DECIDE

Cheapest Is Not Always Best

Before you write your recommendation, check each offer against these security and compliance questions.

Region and residency

Where will the data physically sit, and is that allowed?

Encryption

Is data encrypted at rest and in transit by default?

Identity and access

MFA, roles, and least privilege available out of the box?

Availability (SLA)

What uptime percentage is promised, and what is the remedy?

Certifications

ISO 27001, SOC 2, or similar audits published?

Backup and exit

How easy and how costly is it to back up and leave?

Submit: the filled table + a 5-line recommendation that names at least two security reasons.